Day – 1 Tasks

Part A — User and System Investigation (10 minutes)

Determine the username of the account you are currently using.
: whoami (or) id -un

Display your UID, primary GID, and supplementary groups.
: id (id -u) (id -g) (id -G)

Display all users currently logged in to the system.
: who

Determine what activities the currently logged-in users are performing.
: w

Examine /etc/passwd and identify the following information for your current account:
Username
UID
GID
Home directory
Login shell
: grep "^$(whoami):" /etc/passwd
username:password:UID:GID:gecos:home_directory:login_shell
msis:x:1000:1000:MSIS:/home/msis:/bin/bash

Determine how many user-account entries are present in /etc/passwd.
: wc -l /etc/passwd
57 /etc/passwd

Display only the username and UID fields from /etc/passwd.
: cut -d: -f1,3 /etc/passwd

Find all accounts whose UID is 0. Explain the security significance of UID 0.
: awk -F: '$3 == 0 {print $1}' /etc/passwd
root

Compare /etc/passwd and /etc/shadow. Why are password hashes stored in /etc/shadow instead of /etc/passwd?
: cat /etc/passwd
sudo cat /etc/shadow
ls -l /etc/passwd /etc/shadow

/etc/passwd traditionally needed to be readable by many system programs and users, so storing password hashes there would expose those hashes to every user who could read the file.

/etc/shadow has much more restrictive permissions. This limits access to password hashes and makes offline password-cracking attacks more difficult.

```
msis@worker1:~/Desktop/extglob$ whoami
msis
msis@worker1:~/Desktop/extglob$ id -un
msis
msis@worker1:~/Desktop/extglob$ id
uid=1000(msis) gid=1000(msis) groups=1000(msis),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),100(users),114(lpadmin),125(libvirt)
msis@worker1:~/Desktop/extglob$ who
msis     seat0        2026-09-01 09:05 (login screen)
msis     tty2         2026-09-01 09:05 (tty2)
msis@worker1:~/Desktop/extglob$ users
msis msis
msis@worker1:~/Desktop/extglob$ w
 09:20:19 up 15 min,  1 user,  load average: 0.06, 0.11, 0.09
USER     TTY      FROM             LOGIN@   IDLE   JCPU   PCPU  WHAT
msis     tty2     -                09:05   15:05   0.01s  0.01s /usr/libexec/gnome-session-binary --session=ubuntu
msis@worker1:~/Desktop/extglob$ id -u
1000
msis@worker1:~/Desktop/extglob$ id -g
1000
msis@worker1:~/Desktop/extglob$ id -G
1000 4 24 27 30 46 100 114 125
msis@worker1:~/Desktop/extglob$ id
uid=1000(msis) gid=1000(msis) groups=1000(msis),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),100(users),114(lpadmin),125(libvirt)
msis@worker1:~/Desktop/extglob$ grep "^$(whoami):" /etc/passwd
msis:x:1000:1000:MSIS:/home/msis:/bin/bash
msis@worker1:~/Desktop/extglob$ wc -l /etc/passwd
57 /etc/passwd
msis@worker1:~/Desktop/extglob$ cut -d: -f1,3 /etc/passwd
root:0
daemon:1
```

Part B — Creating and Verifying Users (20 Minutes)

Create the following users and ensure that a home directory is created for each:

alice, bob, charlie

: sudo useradd -m alice
  sudo useradd -m bob
  sudo useradd -m charlie

Verify that all three accounts have been added to /etc/passwd.
: grep -E '^(alice|bob|charlie):' /etc/passwd

Determine the UID assigned to each newly created user.
: id -u alice
  id -u bob
  id -u charlie

Determine the home directory and default shell assigned to alice.
: getent passwd alice

Inspect /home and verify that the required home directories were created.
: ls -la /home
  ls -ld /home/alice /home/bob /home/charlie

Set passwords for all three accounts.
: sudo passwd alice
  sudo passwd bob
  sudo passwd charlie

Switch from your current account to alice using a login shell. Verify the change using an appropriate command, and then return to your original account.

: su – alice
whoami
id
pwd
exit

```
msis@worker1:~/Desktop/extglob$ su - alice
Password:
$ whoami
alice
$ id
uid=1003(alice) gid=1003(alice) groups=1003(alice)
$ pwd
/home/alice
$ echo $SHELL
/bin/sh
$ exit
msis@worker1:~/Desktop/extglob$ whoami
msis
msis@worker1:~/Desktop/extglob$
```

Part C — Group Management and Access Control (30 Minutes)
Create two groups:

security and developers

sudo groupadd security
sudo groupadd developers

verify
getent group security
getent group developers

Add alice and charlie to the security group.
: sudo usermod -aG security alice
sudo usermod -aG security charlie

Add bob to the developers group.
: sudo usermod -aG developers bob

Without switching accounts, verify the group membership of all three users.
: id -Gn alice
id -Gn bob
id -Gn charlie

```
msis@worker1:~/Desktop/extglob$ sudo groupadd security
sudo groupadd developers
msis@worker1:~/Desktop/extglob$ getent group security
getent group developers
security:x:1006:
developers:x:1007:
msis@worker1:~/Desktop/extglob$ sudo usermod -aG security alice
msis@worker1:~/Desktop/extglob$ sudo usermod -aG security charlie
msis@worker1:~/Desktop/extglob$ sudo usermod -aG developers bob
msis@worker1:~/Desktop/extglob$ id -Gn alice
id -Gn bob
id -Gn charlie
alice security
bob developers
charlie security
msis@worker1:~/Desktop/extglob$
```

Display the entries corresponding to security and developers from /etc/group.
: grep -E '^(security|developers):' /etc/group

Format : groupname:password:GID:members
security:x:1006:alice,charlie
developers:x:1007:bob

Add bob to the security group without removing him from the developers group.
: sudo usermod -aG security bob

Verify that bob now belongs to both groups.
: id -Gn bob
bob security developers

```
msis@worker1:~/Desktop/extglob$ id -Gn alice
id -Gn bob
id -Gn charlie
alice security
bob developers
charlie security
msis@worker1:~/Desktop/extglob$ grep -E '^(security|developers):' /etc/group
security:x:1006:alice,charlie
developers:x:1007:bob
msis@worker1:~/Desktop/extglob$ sudo usermod -aG security bob
msis@worker1:~/Desktop/extglob$ id bob
uid=1004(bob) gid=1004(bob) groups=1004(bob),1006(security),1007(developers)
msis@worker1:~/Desktop/extglob$ id -Gn bob
bob security developers
msis@worker1:~/Desktop/extglob$
```